

Exploring the impact of VR and embodiment on environmental literacy

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ABSTRACT

Reduction of global bee populations is one of the relevant topics in the list of environmental issues defined in the United Nations Sustainable Development Goals Agenda. Virtual Reality technologies can play a strategic role in raising awareness and education for sustainability by allowing users to visualize, demonstrate and emphasize information, making it more accessible. Virtual Reality can also elicit emotional involvement, which is important for influencing user behavior. The authors have designed and developed an educational Virtual Reality application called Colonies, which allows users to understand the issues related to the bee problem and empathize with bee colonies. The experience also offers suggestions for more sustainable behaviors.

CCS CONCEPTS

• **Social and professional topics** → Professional topics; Computing industry; Sustainability; • **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality; Interaction design, Ruiyi Liu, Matteo Paoli and Simone Restifo Pilato. 2023. Exploring the impact of VR and embodiment on environmental literacy. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, NumberOfPage <https://doi.org/10.1145/3544549.3583903>.

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1 INTRODUCTION

In 2015, the United Nations defined the Agenda for Sustainable and the related 17 Development Goals (SDGs) [1]. To achieve these goals, many environmental issues must be addressed immediately through a radical change in our daily lives and in the way development has taken place to date. One of these many issues is the reduction in global bee populations [2] Bees are essential for preserving and protecting ecological balance and biodiversity in nature, both for plants and animals, and act as indicators of the state of our environment.

The complexity of the *Bee population* problem requires a systemic approach for being effectively communicated, understood, and addressed. Virtual Reality (VR) technologies can have a strategic role in addressing complex topics, such as that of bees, in the field of awareness and education for sustainability. In fact, they provide a powerful help in visualizing and emphasizing information and in making knowledge more accessible. VR technologies allow making abstract thinking more tangible and elicit users' emotional involvement, which are important aspects for influencing user behaviours. It has been shown that environmental knowledge can drive environmental behavior only if it elicits environmental emotions [3]. Kollmuss et al. reported that emotional involvement shapes environmental awareness and attitude [4] while Dirksen et al. stated that emotional arousal plays an important role in behavioral change and that immersive environments of Extended Reality can result in high emotional arousal [5]. Moreover, several studies show that interactive storytelling, integrated into immersive VR environments, can trigger emotion and empathy, embodied experience, and influence action [6, 7].

On this basis, the authors decided to design and develop an educational interactive VR application named *Colonies* that addresses an important and complex sustainable issue such as that of the *Bee Population*. *Colonies* allows users to understand the issues related

to the bee problem, empathizing with the bee colonies through the user's personal embodiment in a bee during the VR experience. Furthermore, the experience offers some suggestions related to more sustainable behaviors that the user can adopt to counter the problem of bee population reduction. Some preliminary tests were carried out to evaluate the application, which proved to be a useful tool for stimulating motivation and emotional involvement of users and positively influencing their awareness and (possibly) future choices.

2 CONTEXT AND RESEARCH QUESTIONS

Given the significance of sustainability, many scholars have become interested in analysing sustainable behaviour and understanding what influences it. Several studies evaluate and demonstrate that VR can be a powerful tool to help make people more aware and sensitive to sustainable issues. The case of the activist Hossein Ronaghi (2022) [8], in fact, demonstrates that not only VR educational technology is effective on sustainable behaviour, but also that experiences based on VR technology can have a greater effect than traditional training courses on sustainable behaviour. Another interesting study was reported in [9], consisting of a VR experience to teach underwater sustainability, exploiting the different types of empathy that different age groups can have.

A key study was conducted on the impact of the degree of immersion and type of narrative content on environmental behavior [10]. The experiment looks at meat consumption and its effects on ecosystems and shows that people who experience an emotional and engaging video are more likely to actively change their behavior and end up opting for vegetarian food.

One of the most effective methods to make people empathise with sustainable issues and awareness is the sense of embodiment (SoE). Kiltani and Groten (2012) [11] state: "Sense of embodiment is used to refer to the ensemble of sensations that arise in conjunction with being inside, having, and controlling a body, especially in relation to Virtual Reality applications." SoE can be extremely valuable especially when it comes to experiencing an environment, a situation or an entity that is not familiar to the user or that cannot be recreated in real circumstances. SoE in VR applications can be a valuable method to enable users to visualise something that would otherwise be invisible [12]. An example is *Tree*, a virtual tree storytelling experiment where users can feel what it would be like to be another type of life, in this case a tree [13]. The virtual body ownership is applicable for non-humanoids and can even outperform human avatars in some cases, especially when it comes to applications of the embodiment of wild animals for educational and entertainment purposes [14]. It has been widely demonstrated that VR is a new educational tool that can be effectively used in promoting the various facets of environmental literacy among the general population. Furthermore, by exploiting the potential of the SoE, the results can be even more effective. However, no previous study has used VR technology and its relationship to SoE to raise awareness of wild animals, especially pollinators.

The aim of this study is to explore (1) to what extent an immersive VR experience can influence awareness of sustainability and (2) what effect does the experience embodied in VR have on the perception of the sustainable message?

3 DESCRIPTION OF THE INSTALLATION

Colonies is an educational VR application that aims to raise awareness of sustainability and encourage behavioral change. The application demonstrates the importance of connecting all ecosystem players, including pollinators and humans, to ensure a healthy living habitat for the next generation. To increase users' sensitivity and stimulate emotions that can influence action [15], the experience is based on enabling the transformation from human to an embodied bee in an immersive 3D virtual environment. The design of *Colonies* was inspired by *The Camille Stories: Children of Compost* [16] for narrative development and organization. The application consists of four main chapters, plus an introduction and an ending. For each of the four narrative chapters, the story begins with an assumed future time period and an estimated number of existing bees, presented to the user through a Graphical User Interface.

Through the entire storyline, users will follow an emotional interactive narrative that varies from chapter to chapter and is supported by the voice of a narrator throughout the experience. In the first two chapters, the user is focused on learning about sustainable issues related to pollinators in a positive and serene way. Starting at the end of chapter three and in the first part of chapter four, users are exposed to possible negative future conditions and explore adverse environments. To encourage behavioral change, the final part of the application proposes positive scenarios, colors, and audio to motivate people to change their behaviors to help pollinators in their daily lives thanks to practical suggestions.

The application was developed using the Unity 3D game engine (<https://unity.com/>), then exported for the Oculus Meta Quest 2 headset. The first-person view allows users to experience the application individually in one room-scale mode. The virtual environment of *Colonies* consists of a natural forest that simulates the habitat of bees. During the experience, in each chapter it is possible to interact with different elements, using the headset controllers to activate audio sounds and visual feedback. Considering the different levels of confidence of the users in VR environments, and to avoid motion sickness, the authors decided to use teleportation as a means of moving around in the virtual experience. The chapters of the *Colonies* application are presented below, which are: *Introduction*, *Prophecy*, *Metamorphosis*, *Visions*, *Prologue* and *Ending* (Figure 1).

Introduction serves as a tutorial introducing how to interact with controllers and the affordances that will appear in the application. Thanks to visual and text guidance, users of any level of literacy in using similar applications can enjoy the experience without problems.

Prophecy is the beginning of the storyline, where users will enter a peaceful forest scenario. The user is invited to walk on a path in the woods. Along the walking trail, users will be able to interact with different elements, and thanks to a voice narrator, they will begin to understand the imminent problem the bees are facing.

Metamorphosis represents the turning point in the story where the point of view of the human begins to change towards that of the bee. The application presents the different ways in which bees

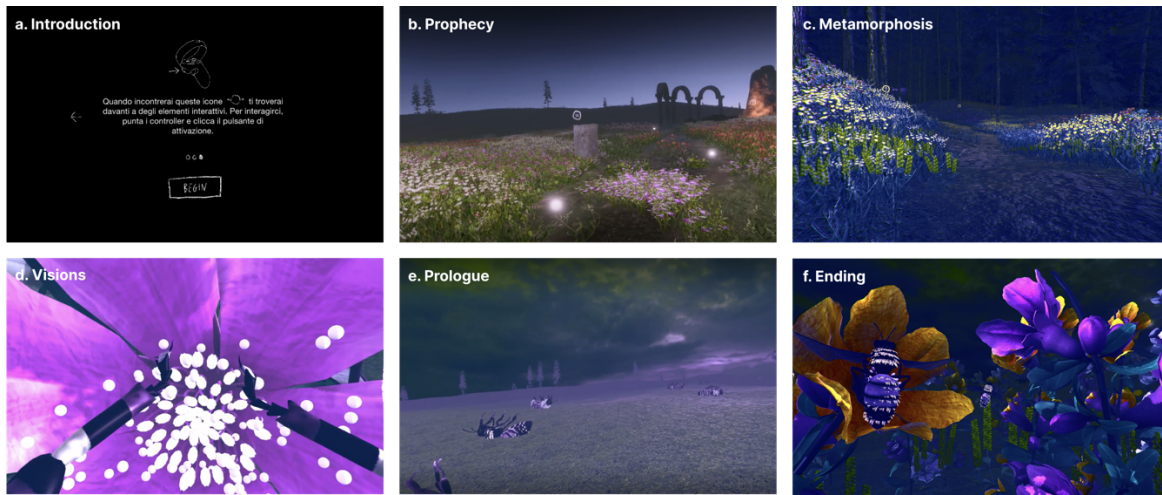


Figure 1: Overview of *Colonies* chapters: (a) Introduction, (b) Prophecy, (c) Metamorphosis, (d) Visions, (e) Prologue, (f) Ending.

and humans perceive colors. Audio narration clearly conveys the experience of transforming the users from human to bee.

Visions is the chapter where the changes in the virtual body and vision will be completed. Users see their VR arms as the paws of bees. Elements of the environment, such as flowers and grass, are magnified to the proportions bees perceive through their eyes. The experience simulates the typical flower-to-flower flight of bees, as well as some interactions with some flowers that trigger pollen release animations.

Prologue represents the culmination of the experience: in a desolate scenario where the bees have become extinct, hope opens up, which holds the final message to be collected. Users enter the chapter seeing the dead bees scattered in the environment and the pollution in the air. However, guided by a vivid live bee companion, users will find themselves faced with a blossoming flower that represents hope. Finally, by selecting an interactive button many more flowers will appear, supported by the audio message which encourages and gives hope to the users and leads to positive emotion.

Ending is the final chapter that aims to advise users to behave more sustainably by providing concrete suggestions that can be easily implemented in daily life, such as buying organic products, planting flowers for pollinators and buying local honey.

4 TESTING SESSION AND DISCUSSION

We collected preliminary results on the *Colonies* VR application. Results involved 10 participants balanced in gender (5 males and 5 females, no other gender has been indicated by participants) between 21 and 56 y.o.. The majority of our sample either had a high school diploma (40%) or a master degree (40%) as the highest level of education. The majority of our participants (6 users) had tried VR a few times, and most of the participants (7 users) defined themselves as “beginner” in the usage of VR instruments. We structured the questionnaire as a first User Experience (UX) pilot to assess the overall level of usability by participants and the level of embodiment with the avatar. The structure of the questionnaire has

taken inspiration from previous works performed by the Mind and Behavior Technological Center (MiBTec) of the University of Milano-Bicocca [17]. We received the approval of the study by the Commission for the Evaluation of the Research of the University of Milano-Bicocca, code RM-2022-597.

4.1 User Experience (UX)

In order to assess the preliminary UX level of the application, we relied on the UX Questionnaire (UEQ-S) in its short form by Schrepp et al. (2017) [18]. We analyzed the relative outcome with the Excel tool provided by the authors of the scale. Results showed an overall good usability ($M = 1.163$, $SD = 1.138$), with satisfactory scores from the Hedonic Quality perspective ($M = 1.250$, $SD = 1.514$) and slightly inferior Pragmatic Quality scores ($M = 1.075$, $SD = 1.242$). The two factors, Hedonic and Pragmatic, should be intended as in Hassenzahl et al. (2003) work [19]: while Pragmatic Quality describes traditional usability elements (e.g., efficiency, effectiveness and learnability) linked to task-related aspects, Hedonic Quality relates to quality elements linked more to the way the user feels in respect of the product (e.g., originality and beauty) [20]. Finally, through a comparison from existing values from a benchmark data set provided by the authors, our applications can be defined as “above average” from the overall perspective, slightly “below average” from the Pragmatic Quality perspective and “good” from the Hedonic Quality point of view.

4.2 Embodiment

We also assessed the level of embodiment experienced by participants in respect of the animal avatar they could impersonate during the final part of the experience. We relied on the Virtual Embodiment Questionnaire (VEQ) [21]. Three main scores have been registered: the level of the Ownership of the body, the level of Agency and the feel of Change in respect of one’s body. We registered an intermediate level of embodiment in Ownership ($M = 4.55$, $SD = 1.27$), Agency ($M = 5.13$, $SD = 0.7$) and Change ($M = 4.17$, $SD = 1.36$).

The Likert scale on which feedback has been registered spanned from 1 (strongly disagree) to 7 (strongly agree).

4.3 Sustainable message perception

We then asked our participants, on a Likert scale from 1 to 7, their level of agreement with the following sentences in respect of the sustainable message transmitted through the VR experience. The questions have been created and customized by the research group. For each phrase we indicate the mean and standard deviation of participants' responses.

- I believe that the application transmits a sustainable message for the protection of bees (M= 6.20, SD= 0.789)
- I believe that the message delivered by the application is clear (M= 6.20, SD= 0.632)
- It has been easy to understand the message delivered by the application (M= 6.20, SD= 0.789)
- I believe that the message delivered by the application is comprehensible for everybody (M= 6.00, SD= 0.816)
- I believe that a Virtual Reality application is a good medium to transmit a sustainable message for the protection of pollinating insects (M= 5.60, SD= 1.51)

From these preliminary results we can say that the message seems to have been correctly delivered to participants, in a clear and understandable way. Further data to confirm or revise these outcomes are needed.

5 CONCLUSIONS

The paper presents a VR application named Colonies that aims to stimulate motivation and emotional engagement of users and to positively influence their awareness and future behaviors toward sustainability. Colonies was developed to raise awareness about declining global bee populations, mainly related to intensive agricultural practices, monocultures, overuse of agricultural chemicals, and higher temperatures associated with climate change (<https://www.fao.org/news/story/en/item/1194910/icode/>). Colonies also presents some suggestions on sustainable actions to be taken in relation to the bee problem, with the aim of motivating the user to change their behaviour. Particular attention has been paid to the design of the UX of the application integrating interactive storytelling to trigger emotions and empathy and embodied experience to influence users' actions. Results of preliminary tests shows that the application is very satisfactory from the general point of view of user experience, that the message of the application is conveyed correctly, and that virtual reality is a good medium to convey sustainable awareness. With regards to the embodiment, the level of embodiment was considered acceptable. In the future, more accurate tests will be conducted to better evaluate the level of user embodiment and better define the success of the VR application in influencing user behavior change and its relationship with the integration of interactive storytelling in VR.

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